

Visual Basic

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History

- Alan Cooper
- Released by Microsoft in 1991
- Evolved from BASIC
- Transitioned to VB.NET in 2002

Why It Was Developed

- Simplify Development
- Rapid Application Development (RAD)

Language Paradigm

Multiparadigm Language

Imperative

- Procedural
- Event-driven programming
 - program can respond to user actions, like clicks, typing, or other events as they happen
- Object-Oriented (full in VB.NET)

Typing System

- Statically Typed
- Weakly Typed

Hello World

```
module h_w
  sUb main()
    Console.WriteLine("hello, world!")
  end sub
end module
```

Output:

hello, world!

Data Types

- Dim ("Dimension") declares variables by setting up a variable's name and type

```
module data_types_example

  sub main()
    'quick data types example, not actual game
    dim greeting as string = "hunt the wumpus"
    dim howmanyarrows as integer = 1
    dim hasarrow as boolean = true

    console.WriteLine("game: " & greeting)
    console.WriteLine("arrows: " & howmanyarrows)
    console.WriteLine("has arrow: " & hasarrow)
  end sub

end module
```

Output:

```
game: hunt the wumpus
arrows: 1
has arrow: True
```

Control Structures

```
module control_structures_example
    function is_palindrome(word as string) as boolean
        'reverses the word and check if it matches original
        dim reversed_word as string = strreverse(word)
        return word = reversed_word
    end function

    sub main()
        'checks multiple words for palindrome
        dim words as string() = {"level", "noon", "visual", "basic"}
        for each word as string in words
            if is_palindrome(word) then
                console.writeline(word & " is a palindrome")
            else
                console.writeline(word & " is not a palindrome")
            end if
        next
    end sub
end module
```

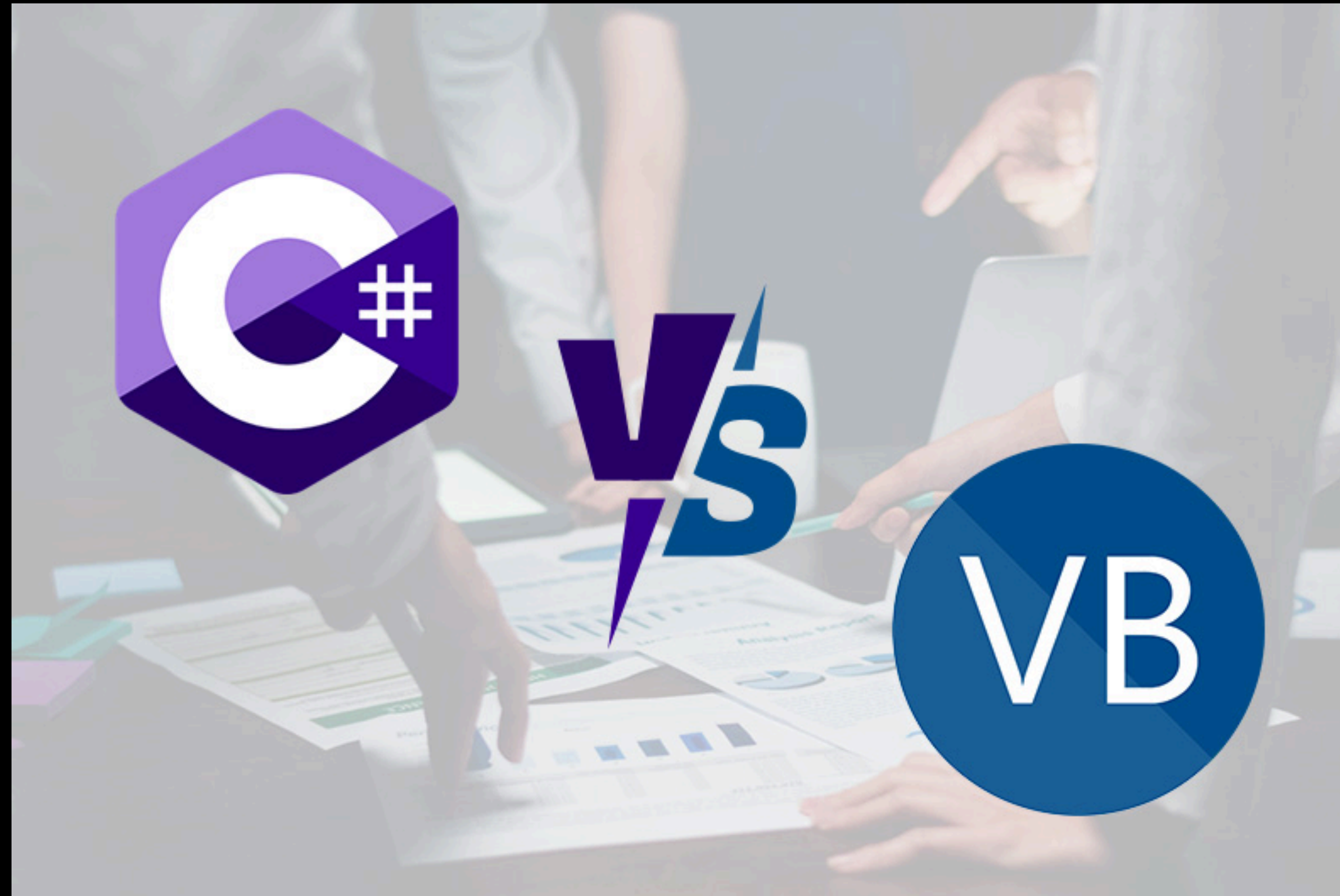
Output:

```
level is a palindrome
noon is a palindrome
visual is not a palindrome
basic is not a palindrome
```


Legacy Languages

- Visual Basic: 1991-1998
- VB.Net: 2002-2019

Comparasion



Problem Domains

- **Business Applications**

examples: accounting, inventory management, and payroll systems

- **Windows-Based Applications**

ideal for creating applications that run on Windows desktops

- **Data-Driven Applications**

works well with databases, like SQL Server, for managing large amounts of data

Features That Make VB Effective

- Strong GUI Support
- Rapid Application Development (RAD)
- Integration with Microsoft Products
- Ease of Learning
- Event-Driven Programming

Questions?